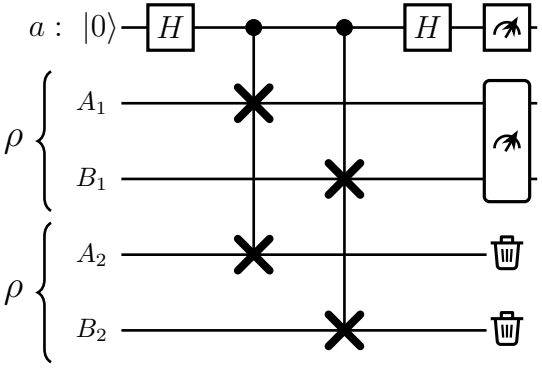


$$\langle O\rangle_{\mathcal{P}} = \frac{\langle Z_a \otimes O_{A_1B_1}\rangle}{\langle Z_a \otimes I\rangle} = \frac{\mathrm{Tr}(O\rho^2)}{\mathrm{Tr}\,\rho^2}$$

Every shot is kept and weighted by the ancilla parity; the state $\rho^2/\,\mathrm{Tr}\,\rho^2$ is never prepared.